Search in course

Search

English

B

1. CS 498: Cloud Computing Applications
2. Week 16
3. MP1 - V2 (Optional - Beta Testing)

PreviousNext

Practice Programming Assignment: MP1 - V2 (Optional - Beta Testing)

You have not submitted.

Deadline

Pass this assignment by May 11, 11:59 PM CDT

InstructionsMy submissionsDiscussions

MP1 v2 - EC2 and S3 (OPTIONAL)

Overview

This MP is designed to teach the students about EC2, S3 and IAM roles and policies needed to set up a working flow between these technologies.

In this assignment, you will launch an EC2 instance, configure it with an IAM role that has S3 access, and write a Python script to process CSV data from S3. The objective is to download a CSV file of orders from a public S3 bucket, filter the data based on specific conditions, and then upload the filtered results to your own S3 bucket. Finally, you'll run a submission script that will verify your setup and your filtered results.

Requirements

Before you begin, ensure you have the following:

- **AWS Account:** Access to the AWS Console.
- **EC2 Instance:** Launch an EC2 instance using Amazon Linux 2 or Ubuntu (t2.micro for free-tier eligibility).
- **IAM Role:** Create and attach an IAM role with the following policies:
    - AmazonS3FullAccess
    - AmazonEC2ReadOnlyAccess
    - AmazonSSMManagedInstanceCore
    - IAMReadOnlyAccess
- **S3 Bucket:**
    - One personal S3 bucket where you will upload your processed file.

Procedure

Section 1: Setup EC2

Step 1.1 - Launch EC2

- Launch one EC2 instance. We suggest using the latest Amazon Linux AMI as the operating system and the smallest EC2 instance possible (e.g., Nano or Micro) to make the cost of your instances as small as possible (or free.) Refer to the link below for the specific steps.
    - https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EC2_GetStarted.html
- Ensure you can connect to the EC2 instances using terminal (or the embedded terminal in the EC2 console on your browser) via ssh.
- Install Python on the EC2 machine.

Step 1.2 - Create IAM Role

Create an IAM role with the policies listed in the requirements. You need to create a AWS Service Type role for EC2 to access permissions.

Relevant Documentation -

- https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles.html

- https://docs.aws.amazon.com/IAM/latest/UserGuide/id_roles_create_for-service.html

Step 1.3 - Attach the role to the EC2 instance

We need the EC2 instance to have the permissions defined in the IAM role so we need to attach the IAM role to the EC2 instance. Follow the steps below to achieve this.

Documentation - https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/attach-iam-role.html

Section 2: Process Data

Step 2.1 - Download the CSV file

Use the AWS CLI or Python's boto3 to download orders.csv from the public S3 bucket and on your EC2 machine.

Bucket - s3://mp1-orders/orders.csv

Refer to this documentation - https://docs.aws.amazon.com/cli/latest/reference/s3/

Step 2.2 - Filter the Data

The orders.csv contains the following columns, OrderID,CustomerID,ProductName,Category,Price,OrderDate,Quantity,City,PaymentMethod,OrderStatus. Write a **Python Script** to filter out all rows and store this in a file called results.csv on your EC2 machine.

- Category is Electronics
- Price is greater than 300
- Payment method is PayPal

Section 3: Create your S3 Bucket (Public)

To upload your processed file, you'll need to create your own S3 bucket. Follow these steps:

1. **Create an S3 Bucket:**Log in to the AWS Console and navigate to **S3**.
    - Click **Create bucket**.
    - Provide a unique bucket name.
    - **Uncheck Block all public access** and confirm that you want to make the bucket public. *(Note: Making a bucket public means that anyone on the*

internet may have read access to its contents. Ensure you do this only for non-sensitive data and delete the bucket after the MP is completed.)

- Click **Create bucket**.

2. **Configure Public Access:**
    - Add a bucket policy to allow public read access to objects in the bucket.
    - Under permissions, Edit the following policy and add it to your bucket!

```
{
 "Version": "2012-10-17",
 "Statement": [
  {
   "Sid": "PublicReadGetObject",
   "Effect": "Allow",
   "Principal": "*",
   "Action": "s3:GetObject",
```

```
    "Resource": "arn:aws:s3:::<your-bucket-name>/*"
  }
 ]
}
```

- Replace <your-bucket-name> with your actual bucket name and save the policy.

Section 4: Upload your results.csv to your own S3 bucket

Similar to Section 2, upload your results.csv file to the bucket and copy the URI for your file. It should look like this “s3://<your-s3-bucket>/results.csv”

Section 5: Submission

Copy the **submit.py** script given here on your EC2 machine.

submit

PY File

python3 submit.py --coursera-token YOUR_COURERA_TOKEN --s3-uri s3://your-bucket/your-file --submitter-email YOUR EMAIL

You will receive the autograder's feedback for the current submission, and your grade on Coursera will be updated automatically. **Please TERMINATE all EC2 instances, EMPTY and DELETE S3 bucket after the MP to avoid incurring charges!**

FAQ

Q1: How can I verify that my file was correctly uploaded to my S3 bucket?

A1: Use the AWS CLI command aws s3 ls s3://<your-s3-bucket>/ to list the files in your bucket and confirm that your file is present.

How to submit

Copy the token below and run the submission script included in the assignment download. When prompted, use your email address **fridkis2@illinois.edu**.

wqsTV950BeqCoOLj

Generate new token

Your submission token is unique to you and should not be shared with anyone. You may submit as many times as you like.

Like

Dislike

Report an issue